

Deepak Mallampati

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Kent, OH (US-based, F-1 OPT, open to sponsorship)

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) who ships customer-facing agentic systems end-to-end. Production browser-extension (Manga Lens, Chrome Web Store, MV3 + multi-provider LLM + 29-site selector configs), healthcare RAG with ~35% retrieval precision gains, and multi-agent claims pipelines with schema-validated JSON contracts. Master's degree (Kent State). Founding-engineer energy with HIPAA-conscious production discipline.

CORE COMPETENCIES

Forward Deployed / Founding Engineer • Agentic Workflows + LLM Orchestration • Browser Automation (Chrome MV3, content scripts, service workers) • Multi-provider LLM (OpenAI, Anthropic, Gemini, Ollama) • RAG + Embeddings + Vector Search • Customer-facing Delivery + Stakeholder Comms • Python + TypeScript + FastAPI + Docker • QA, Debugging, Reliability

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee

USA | Healthcare Technology, AI Consulting, SaaS Healthcare

- Shipped **agentic LLM workflows** for multi-step healthcare queries (eligibility, care navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; improved agent response stability by ~25%.
- Built **Retrieval-Augmented Generation (RAG)** for clinical knowledge retrieval; recursive semantic chunking + transformer embeddings improved retrieval precision by ~35%; retrieval-grounded response alignment exceeded 90%.
- Owned customer-facing stakeholder communication: system-limitation docs, evaluation audit trails, de-identification gates - HIPAA-conscious governance with clean clinical sign-off.
- Packaged ML/LLM inference as **FastAPI/Flask + Docker** with structured logging and load simulation; reduced post-deployment defects by ~30%.
- Predictive ML pipelines (scikit-learn, XGBoost) for patient no-show + care engagement; **15-20% recall** gains on high-risk categories with precision >90%.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

Hyderabad, India | Oil & Gas, Enterprise ERP

- Optimized T-SQL stored procedures (Synthesis Order-to-Cash); reduced query execution time ~20% and data retrieval latency ~25%.
- Built Jenkins CI/CD pipelines for schema + stored-procedure deployments; reduced deployment errors >30% and improved release cycle time ~35-40%.
- Built SQL Server DMV + Grafana dashboards for long-running queries, deadlocks, CPU/I/O contention; cut incident recurrence ~25%.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer

Hyderabad, India | Nonprofit Tech, Video Analytics, Applied NLP

- Built transformer-based video summarization + highlight-generation across **5,000+ recorded sessions**; reduced manual review time **60-70%** and achieved ~85% alignment with human-curated key moments.
- Document Q&A with RAG (semantic chunking + embedding retrieval); ~30% hallucination reduction; >85% contextual answer accuracy.
- Deployed as a Flask API with a lightweight web UI for non-technical staff.

PROJECTS

Manga Lens - Chrome Extension for Real-Time AI Manga/Webtoon Translation shipped, Chrome Web Store

Solo full-stack Chrome MV3 extension. Content scripts, service workers, captureVisibleTab for panel capture. Integrates **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v local). Multi-section viewport-slice capture pipeline with coordinate remap + dedup. Per-domain selector configs across **29 manga/webtoon sites**, seven-day translation cache, narrowed host_permissions + privacy policy. Multi-provider payload handling (WebP cloud, JPEG for Ollama to avoid CUDA crash).

Stack: Chrome MV3, TypeScript, content scripts, service workers, multi-provider LLM orchestration

Agentic Pixel Character Synthesis & Animation Engine ongoing

Phased generative AI for identity-consistent, pose-controlled pixel characters with sprite-sheet export. **LLM-based agent orchestrator** decomposes high-level prompts into generation tasks (SD fine-tune + LoRA identity, ControlNet + OpenPose pose, latent-space consistency for animation).

Stack: PyTorch, Hugging Face Diffusers, FastAPI, Docker, LLM orchestration

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake → Validation → Consistency → Duplicate → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; explainable risk scoring with audit-ready reasoning traces.

Dream Decoder - AI-Powered Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app coordinating multimodal APIs. Introduced **intermediate structured prompt transformation layers** - improved contextual alignment ~30% and first-pass image success rate ~25-30%.

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

WORK AUTHORIZATION

F-1 OPT

- authorized to work in the United States now; future sponsorship required.

SKILLS

Languages & Frameworks: Python, TypeScript, FastAPI, Flask, React, SQL (T-SQL, PostgreSQL, SQLite), C++ (Arduino), PHP

LLM & Agentic: Multi-provider LLM orchestration, RAG, agentic workflows, structured outputs, prompt engineering, tool calling, evaluation pipelines, guardrails, grounding, embeddings, vector search

Browser & Web Automation: Chrome MV3, content scripts, service workers, captureVisibleTab, per-domain selector configs, host_permissions narrowing, multi-provider payload handling

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers, LangChain, LlamaIndex

Computer Vision & Multimodal: YOLOv8, OpenCV, ControlNet, OpenPose/MediaPipe, Stable Diffusion

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, observability/logging

Domains: Forward Deployed Engineering, Applied AI, Browser Agents, Healthcare AI (HIPAA-conscious), Workflow Automation